Panskura Banamali College (Autonomous)

B.Sc.(H) 5th Semester Examination 2022 (Online mode)

Subject : Computer Science Paper: CC12T

Paper Name : Theory of Computation

F.M. : 40 Time : 2 Hours

Answer any four questions : 4×10=40

1. a) Design a machine (DFA) which checks whether a given decimal number is even.

   b) What is the difference between DFA & NFA.

   (5+5)

2. a) Define the following terms
   i) Concatination
   ii) Deterministic finite automata (DFA)
   iii) Non-deterministic automata (DFA)
   iv) ε –closure.
   v) Kleen closure & Positive closure

   5×2=10

3. a) Check whether given language id CFL? Or not $L=\{a^m b^{2m} a^m \mid m>=1\}$

   b) Convert the following grammar into CNF
   S->aAab|Aba
   A->aS|bB
   B->ASb|a

   (5+5)

4. a) Construct NFA for accepting string over {a, b} starts with 'a' and ends with 'b'.
   b) Find out regular expression of the language consisting of all string over {0, 1} with at least two consecutive 0's.

   (5+5)

5. a) Write a LEX program to find out the factorial of a given number.
   b) What are the functions of lexical analyser? What is input buffering? Differentiate between Programs and application program.

   (5+2+3)

6. a) What are tokens? Describe different types several symbols involved to represent regular Expression.
   b) Write a LEX program which recognize English words according to English grammar

   (5+5)

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